Near-Field Calibration of MIMO Radar with Misalignment Estimation and Compensation

Xin Zhang, Zhengzheng Li, Shuimei Zhang, Yu Zhang

I. Near-Field Calibration

Near-field calibration owns several benefits comparing with far-field calibration. First, it can reduce the chamber size in our manufacturing plants, which will further save physical space and financial cost. Second, it can increase our sensor test capability. For example, if a 7-meter chamber is replaced by three 2-meter chambers, twice more radar units can be calibrated simultaneously. Last but not least, near-field calibration can support mass production of modern automotive radar with large aperture. As the autonomous driving technologies evolve, the need of radar with very fine angular resolution is on the rise. Imaging radar with larger and larger aperture is the mainstream direction, while classical far-field assumption in the chamber fails when it comes to these modern automotive radar units. Therefore, studying near-field calibration will surely help future mass production of our sensors.

Implementing near-field calibration faces challenges, however. First, the reference steering vectors are different from the counterparts in far-field calibration. As shown in Fig. 1, the far-field wavefronts are parallel, and the phase difference is range independent. The steering vectors can be specified on inter-element spacing and arrival angle. In Fig. 2, we can see that the near-field wavefronts are not

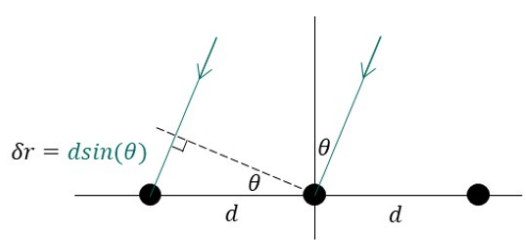

Fig. 1: Far-field model with a uniform linear array.

parallel, and the phase difference is range dependent. Specifically, we need to know the target, receiver, and transmitter locations in order to generate the steering vectors. The second challenge is that, to obtain accurate relative positions among target, receiver, and transmitter, the misalignment of the radar board

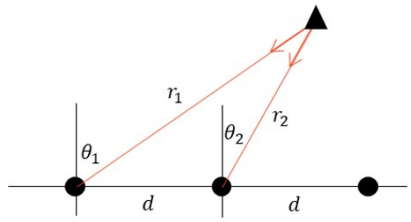

Fig. 2: Near-field model with a uniform linear array.

should be taken into account. This misalignment could be introduced by uneven screwing in the radar board housing, mounting error during attaching the sensor to the rotator, or any mechanical looseness in the system. Without considering the misalignment, errors caused by model mismatch will harm the calibration performance.

Different from far-field calibration where only roll-pitch-yaw misalignment is involved, near-field misalignment will consider rotation center offset and roll-pitch-yaw misalignment. An ideal chamber setup is shown in Fig. 3, where the rotator coordinate system overlaps with radar board coordinate system and the rotation center overlaps with the coordinate origin of the radar board. In practice, the

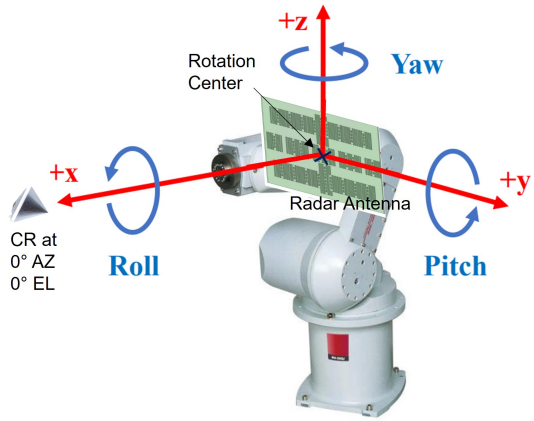

Fig. 3: Chamber setup without misalignment, i.e., ideal case.

near-field misalignment is inevitable, and such misalignment cannot be accurately measured by ruler or laser in our chamber. As shown in Fig. 4, the often-seen case is that the rotator boresight and radar board boresight are pointing to the different directions, and usually there is an offset between the rotation center and the radar board coordinate origin. A small amount of such misalignment can degrade the calibration performance a lot under our current evaluation framework. In the next section, we will model the near-field data with the misalignment mentioned above.

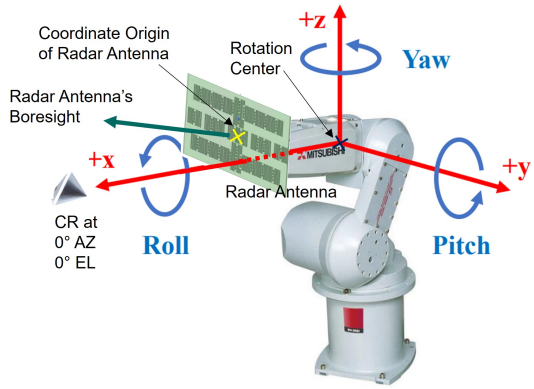

Fig. 4: Chamber setup with misalignment, i.e., real case.

II. Near-Field Data Model

Consider an antenna board with M transmitters and N receivers in the near-field model, and T angles (targets) are swept upon to collect calibration data. Thus, we have T beamvectors (BVs), and each BV has $M \times N$ elements. The $\{m, n, t\}$-th element in the steering matrix can be written as

$$\phi_{m,n,t} = \exp\left\{ \jmath 2\pi \frac{d_{m,n,t}}{\lambda} \right\}, \tag{1}$$

where $d_{m,n,t}$ is the path length from the m-th transmitter to the t-th target, and back to the n-th receiver, i.e.,

$$d_{m,n,t} = d_{m,t} + d_{n,t}. \tag{2}$$

As shown in Fig. 5, the distance between the m-th transmitter and the t-th target is $d_{m,t} = \|\mathbf{p}_m - \mathbf{q}_t\|$,

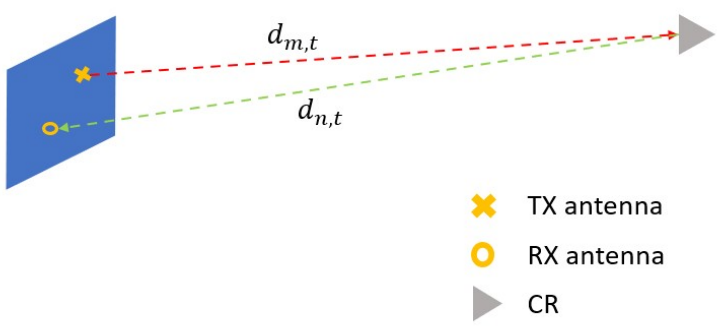

Fig. 5: The transmission path of the m-th transmitter, the t-th target, and the n-th receiver.

and the distance between the n-th receiver and the t-th target is $d_{n,t} = \|\mathbf{r}_n - \mathbf{q}_t\|$, where $\mathbf{p}_m$, $\mathbf{r}_n$, and

$\mathbf{q}_t$ are the Cartesian coordinates of the m-th transmitter, n-th receiver, and t-th target, respectively. Taking the roll-pitch-yaw misalignment and rotating center offset into account, we have

$$\begin{aligned}\mathbf{p}_m =& \mathbf{R}\mathbf{p}_m^o + \boldsymbol{\delta} \\ \mathbf{r}_n =& \mathbf{R}\mathbf{r}_n^o + \boldsymbol{\delta},\end{aligned} \tag{3}$$

where $\mathbf{R}$ is the rotating matrix depending on roll-pitch-yaw misalignment angles $\mathbf{v} = [\gamma, \beta, \alpha]^T$, $\mathbf{p}_m^o$ is the original coordinates of the m-th transmitter, $\mathbf{r}_n^o$ is the original coordinates of the n-th receiver, and $\boldsymbol{\delta} = [\delta_x, \delta_y, \delta_z]^T$ is the rotating center offset. In the above model, the unknowns are $\mathbf{R}$ and $\boldsymbol{\delta}$, while $\{\mathbf{p}_m^o\}$, $\{\mathbf{r}_n^o\}$, and $\{\mathbf{q}_t\}$ are all known. The rotating matrix $\mathbf{R}$ can be written as a function of roll-pitch-yaw misalignment angles as

$$\mathbf{R} = \begin{bmatrix} \cos\beta\cos\alpha & \cos\beta\sin\alpha & -\sin\beta \\ \sin\gamma\sin\beta\cos\alpha - \cos\gamma\sin\alpha & \sin\gamma\sin\beta\sin\alpha + \cos\gamma\cos\alpha & \sin\gamma\cos\beta \\ \cos\gamma\sin\beta\cos\alpha + \sin\gamma\sin\alpha & \cos\gamma\sin\beta\sin\alpha - \sin\gamma\cos\alpha & \cos\gamma\cos\beta \end{bmatrix}. \tag{4}$$

Before visiting the cost function associated with our calibration problem, we would like to prepare some equations which will be used to derive the gradient vector and Hessian matrix in the coming optimization problem.

III. Prerequisite Derivatives

In this section, we will present some partial derivatives that are useful in our later derivations. First, the first partial derivatives of the path length with respect to roll-pitch-yaw misalignment angles are developed. Then, the first partial derivatives with respect to both misalignment angles and rotating center offsets are developed. Lastly, the second partial derivatives are derived.

A. First partial derivatives w.r.t. misalignment angles

At this stage, suppose we only have the roll-pitch-yaw misalignment angles $\mathbf{v} = [\gamma, \beta, \alpha]^T$ as the unknowns (e.g., $\boldsymbol{\delta} = \mathbf{0}$), and we have

$$\frac{\partial d_{m,n,t}}{\partial \mathbf{v}} = \frac{\partial d_{m,t}}{\partial \mathbf{v}} + \frac{\partial d_{n,t}}{\partial \mathbf{v}} = \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \mathbf{v}} + \frac{\partial \|\mathbf{r}_n - \mathbf{q}_t\|}{\partial \mathbf{v}}. \tag{5}$$

To elaborate the above equations, we have derivatives

$$\frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \mathbf{v}} = \frac{\mathbf{J}_m^T (\mathbf{p}_m - \mathbf{q}_t)}{\|\mathbf{p}_m - \mathbf{q}_t\|}, \tag{6}$$

and

$$\frac{\partial \|\mathbf{r}_n - \mathbf{q}_t\|}{\partial \mathbf{v}} = \frac{\mathbf{K}_n^T (\mathbf{r}_n - \mathbf{q}_t)}{\|\mathbf{r}_n - \mathbf{q}_t\|}, \tag{7}$$

where the Jacobian matrices are

$$\mathbf{J}_m = \frac{\partial\left(\mathbf{p}_m - \mathbf{q}_t\right)}{\partial \mathbf{v}} = \left[\frac{\partial\left(\mathbf{p}_m - \mathbf{q}_t\right)}{\partial \gamma}, \frac{\partial\left(\mathbf{p}_m - \mathbf{q}_t\right)}{\partial \beta}, \frac{\partial\left(\mathbf{p}_m - \mathbf{q}_t\right)}{\partial \alpha}\right] = \left[\frac{\partial \mathbf{R}}{\partial \gamma}\mathbf{p}_m^o, \frac{\partial \mathbf{R}}{\partial \beta}\mathbf{p}_m^o, \frac{\partial \mathbf{R}}{\partial \alpha}\mathbf{p}_m^o\right], \quad (8)$$

and

$$\mathbf{K}_n = \frac{\partial\left(\mathbf{r}_n - \mathbf{q}_t\right)}{\partial \mathbf{v}} = \left[\frac{\partial \mathbf{R}}{\partial \gamma}\mathbf{r}_n^o, \frac{\partial \mathbf{R}}{\partial \beta}\mathbf{r}_n^o, \frac{\partial \mathbf{R}}{\partial \alpha}\mathbf{r}_n^o\right]. \quad (9)$$

The first partial derivatives of $\mathbf{R}$ are given by

$$\frac{\partial \mathbf{R}}{\partial \gamma} = \begin{bmatrix} 0 & 0 & 0 \\ \cos\gamma\sin\beta\cos\alpha + \sin\gamma\sin\alpha & \cos\gamma\sin\beta\sin\alpha - \sin\gamma\cos\alpha & \cos\gamma\cos\beta \\ -\sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha & -\sin\gamma\sin\beta\sin\alpha - \cos\gamma\cos\alpha & -\sin\gamma\cos\beta \end{bmatrix}, \quad (10)$$

$$\frac{\partial \mathbf{R}}{\partial \beta} = \begin{bmatrix} -\sin\beta\cos\alpha & -\sin\beta\sin\alpha & -\cos\beta \\ \sin\gamma\cos\beta\cos\alpha & \sin\gamma\cos\beta\sin\alpha & -\sin\gamma\sin\beta \\ \cos\gamma\cos\beta\cos\alpha & \cos\gamma\cos\beta\sin\alpha & -\cos\gamma\sin\beta \end{bmatrix}, \quad (11)$$

$$\frac{\partial \mathbf{R}}{\partial \alpha} = \begin{bmatrix} -\cos\beta\sin\alpha & \cos\beta\cos\alpha & 0 \\ -\sin\gamma\sin\beta\sin\alpha - \cos\gamma\cos\alpha & \sin\gamma\sin\beta\cos\alpha - \cos\gamma\sin\alpha & 0 \\ -\cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & \cos\gamma\sin\beta\cos\alpha + \sin\gamma\sin\alpha & 0 \end{bmatrix}. \quad (12)$$

B. First partial derivatives w.r.t. misalignment angles and rotating center offsets

Suppose we have the roll-pitch-yaw misalignment angles and rotating center offsets as the unknowns $\boldsymbol{\nu} = [\nu_1, \nu_2, \nu_3, \nu_4, \nu_5, \nu_6]^T = [\gamma, \beta, \alpha, \delta_x, \delta_y, \delta_z]^T$, and the first derivatives can be written as

$$\frac{\partial d_{m,n,t}}{\partial \boldsymbol{\nu}} = \frac{\partial d_{m,t}}{\partial \boldsymbol{\nu}} + \frac{\partial d_{n,t}}{\partial \boldsymbol{\nu}}. \quad (13)$$

Specifically, we have the following derivatives

$$\frac{\partial d_{m,t}}{\partial \boldsymbol{\nu}} = \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \boldsymbol{\nu}} = \mathbf{J}_m^T \mathbf{b}_{m,t}, \quad (14)$$

$$\frac{\partial d_{n,t}}{\partial \boldsymbol{\nu}} = \frac{\partial \|\mathbf{r}_n - \mathbf{q}_t\|}{\partial \boldsymbol{\nu}} = \mathbf{K}_n^T \mathbf{c}_{n,t}, \quad (15)$$

where $\mathbf{b}_{m,t} = \frac{\mathbf{p}_m - \mathbf{q}_t}{\|\mathbf{p}_m - \mathbf{q}_t\|}$, $\mathbf{c}_{n,t} = \frac{\mathbf{r}_n - \mathbf{q}_t}{\|\mathbf{r}_n - \mathbf{q}_t\|}$, and

$$\mathbf{J}_m = \frac{\partial\left(\mathbf{p}_m - \mathbf{q}_t\right)}{\partial \boldsymbol{\nu}} = \left[\frac{\partial \mathbf{R}}{\partial \gamma}\mathbf{p}_m^o, \frac{\partial \mathbf{R}}{\partial \beta}\mathbf{p}_m^o, \frac{\partial \mathbf{R}}{\partial \alpha}\mathbf{p}_m^o, \mathbf{I}_3\right], \quad (16)$$

$$\mathbf{K}_n = \frac{\partial\left(\mathbf{r}_n - \mathbf{q}_t\right)}{\partial \boldsymbol{\nu}} = \left[\frac{\partial \mathbf{R}}{\partial \gamma}\mathbf{r}_n^o, \frac{\partial \mathbf{R}}{\partial \beta}\mathbf{r}_n^o, \frac{\partial \mathbf{R}}{\partial \alpha}\mathbf{r}_n^o, \mathbf{I}_3\right], \quad (17)$$

with $\mathbf{I}_3$ as the 3×3 identity matrix. To prepare more results for our future derivations, we obtain the first parital derivatives of $\mathbf{b}_{m,t}$ as follows,

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \gamma} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left(\frac{\partial \mathbf{R}}{\partial \gamma} \mathbf{p}_m^o - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \gamma} \right), \tag{18}$$

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \beta} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left(\frac{\partial \mathbf{R}}{\partial \beta} \mathbf{p}_m^o - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \beta} \right), \tag{19}$$

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \alpha} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left(\frac{\partial \mathbf{R}}{\partial \alpha} \mathbf{p}_m^o - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \alpha} \right), \tag{20}$$

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \delta_x} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left([1,0,0]^T - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \delta_x} \right), \tag{21}$$

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \delta_y} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left([0,1,0]^T - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \delta_y} \right), \tag{22}$$

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \delta_z} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left([0,0,1]^T - \mathbf{b}_{m,t} \frac{\partial \|\mathbf{p}_m - \mathbf{q}_t\|}{\partial \delta_z} \right). \tag{23}$$

With some manipulation, we get

$$\frac{\partial \mathbf{b}_{m,t}}{\partial \boldsymbol{\nu}} = \frac{1}{\|\mathbf{p}_m - \mathbf{q}_t\|} \left(\mathbf{I}_3 - \mathbf{b}_{m,t} \mathbf{b}_{m,t}^T \right) \mathbf{J}_m. \tag{24}$$

Similarly and obviously, we have the result for $\mathbf{c}_{n,t}$, i.e.,

$$\frac{\partial \mathbf{c}_{n,t}}{\partial \boldsymbol{\nu}} = \frac{1}{\|\mathbf{r}_n - \mathbf{q}_t\|} \left(\mathbf{I}_3 - \mathbf{c}_{n,t} \mathbf{c}_{n,t}^T \right) \mathbf{K}_n. \tag{25}$$

C. Second partial derivatives w.r.t. misalignment angles and rotating center offsets

For $t = 1, 2, ..., T$, the second derivatives can be written as

$$\frac{\partial^2 d_{m,n,t}}{\partial \boldsymbol{\nu}^2} = \frac{\partial^2 d_{m,t}}{\partial \boldsymbol{\nu}^2} + \frac{\partial^2 d_{n,t}}{\partial \boldsymbol{\nu}^2}, \tag{26}$$

and, by noting the results in (14)–(17), we have

$$\frac{\partial^2 d_{m,t}}{\partial \boldsymbol{\nu}^2} = \frac{\partial (\mathbf{J}_m^T \mathbf{b}_{m,t})}{\partial \boldsymbol{\nu}} = \left[\frac{\partial \mathbf{J}_m^T}{\partial \gamma} \mathbf{b}_{m,t}, \frac{\partial \mathbf{J}_m^T}{\partial \beta} \mathbf{b}_{m,t}, \frac{\partial \mathbf{J}_m^T}{\partial \alpha} \mathbf{b}_{m,t}, \mathbf{0}_{6\times 3} \right] + \mathbf{J}_m^T \frac{\partial \mathbf{b}_{m,t}}{\partial \boldsymbol{\nu}}, \tag{27}$$

and

$$\frac{\partial^2 d_{n,t}}{\partial \boldsymbol{\nu}^2} = \frac{\partial (\mathbf{K}_n^T \mathbf{c}_{n,t})}{\partial \boldsymbol{\nu}} = \left[\frac{\partial \mathbf{K}_n^T}{\partial \gamma} \mathbf{c}_{n,t}, \frac{\partial \mathbf{K}_n^T}{\partial \beta} \mathbf{c}_{n,t}, \frac{\partial \mathbf{K}_n^T}{\partial \alpha} \mathbf{c}_{n,t}, \mathbf{0}_{6\times 3} \right] + \mathbf{K}_n^T \frac{\partial \mathbf{c}_{n,t}}{\partial \boldsymbol{\nu}}. \tag{28}$$

Based on the results in (16) and (17), the partial derivatives of the Jacobian matrices are

$$\frac{\partial \mathbf{J}_m}{\partial \gamma} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma^2} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \beta \partial \gamma} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \gamma} \mathbf{p}_m^o, \mathbf{0}_{3\times 3} \right], \tag{29}$$

$$\frac{\partial \mathbf{J}_m}{\partial \beta} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \beta} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \beta^2} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \beta} \mathbf{p}_m^o, \mathbf{0}_{3\times 3} \right], \tag{30}$$

$$\frac{\partial \mathbf{J}_m}{\partial \alpha} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \alpha} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \beta \partial \alpha} \mathbf{p}_m^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha^2} \mathbf{p}_m^o, \mathbf{0}_{3\times 3} \right], \tag{31}$$

$$\frac{\partial \mathbf{K}_n}{\partial \gamma} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma^2} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \beta \partial \gamma} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \gamma} \mathbf{r}_n^o, \mathbf{0}_{3\times 3} \right], \tag{32}$$

$$\frac{\partial \mathbf{K}_n}{\partial \beta} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \beta} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \beta^2} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \beta} \mathbf{r}_n^o, \mathbf{0}_{3\times 3} \right], \tag{33}$$

$$\frac{\partial \mathbf{K}_n}{\partial \alpha} = \left[\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \alpha} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \beta \partial \alpha} \mathbf{r}_n^o, \frac{\partial^2 \mathbf{R}}{\partial \alpha^2} \mathbf{r}_n^o, \mathbf{0}_{3\times 3} \right], \tag{34}$$

$$\frac{\partial \mathbf{J}_m}{\partial \delta_x} = \frac{\partial \mathbf{J}_m}{\partial \delta_y} = \frac{\partial \mathbf{J}_m}{\partial \delta_z} = \frac{\partial \mathbf{K}_n}{\partial \delta_x} = \frac{\partial \mathbf{K}_n}{\partial \delta_y} = \frac{\partial \mathbf{K}_n}{\partial \delta_z} = \mathbf{0}_{3\times 6}. \tag{35}$$

Based on the results in (10)–(12), the second partial derivatives of $\mathbf{R}$ are shown as

$$\frac{\partial^2 \mathbf{R}}{\partial \gamma^2} = \begin{bmatrix} 0 & 0 & 0 \\ -\sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha & -\sin\gamma\sin\beta\sin\alpha - \cos\gamma\cos\alpha & -\sin\gamma\cos\beta \\ -\cos\gamma\sin\beta\cos\alpha - \sin\gamma\sin\alpha & -\cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & -\cos\gamma\cos\beta \end{bmatrix}, \tag{36}$$

$$\frac{\partial^2 \mathbf{R}}{\partial \beta^2} = \begin{bmatrix} -\cos\beta\cos\alpha & -\cos\beta\sin\alpha & \sin\beta \\ -\sin\gamma\sin\beta\cos\alpha & -\sin\gamma\sin\beta\sin\alpha & -\sin\gamma\cos\beta \\ -\cos\gamma\sin\beta\cos\alpha & -\cos\gamma\sin\beta\sin\alpha & -\cos\gamma\cos\beta \end{bmatrix}, \tag{37}$$

$$\frac{\partial^2 \mathbf{R}}{\partial \alpha^2} = \begin{bmatrix} -\cos\beta\cos\alpha & -\cos\beta\sin\alpha & 0 \\ -\sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha & -\sin\gamma\sin\beta\sin\alpha - \cos\gamma\cos\alpha & 0 \\ -\cos\gamma\sin\beta\cos\alpha - \sin\gamma\sin\alpha & -\cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & 0 \end{bmatrix}, \tag{38}$$

$$\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \beta} = \frac{\partial^2 \mathbf{R}}{\partial \beta \partial \gamma} = \begin{bmatrix} 0 & 0 & 0 \\ \cos\gamma\cos\beta\cos\alpha & \cos\gamma\cos\beta\sin\alpha & -\cos\gamma\sin\beta \\ -\sin\gamma\cos\beta\cos\alpha & -\sin\gamma\cos\beta\sin\alpha & \sin\gamma\sin\beta \end{bmatrix}, \tag{39}$$

$$\frac{\partial^2 \mathbf{R}}{\partial \gamma \partial \alpha} = \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \gamma} = \begin{bmatrix} 0 & 0 & 0 \\ -\cos\gamma\sin\beta\sin\alpha + \sin\gamma\cos\alpha & \cos\gamma\sin\beta\cos\alpha + \sin\gamma\sin\alpha & 0 \\ \sin\gamma\sin\beta\sin\alpha + \cos\gamma\cos\alpha & -\sin\gamma\sin\beta\cos\alpha + \cos\gamma\sin\alpha & 0 \end{bmatrix}, \tag{40}$$

$$\frac{\partial^2 \mathbf{R}}{\partial \beta \partial \alpha} = \frac{\partial^2 \mathbf{R}}{\partial \alpha \partial \beta} = \begin{bmatrix} \sin\beta\sin\alpha & -\sin\beta\cos\alpha & 0 \\ -\sin\gamma\cos\beta\sin\alpha & \sin\gamma\cos\beta\cos\alpha & 0 \\ -\cos\gamma\cos\beta\sin\alpha & \cos\gamma\cos\beta\cos\alpha & 0 \end{bmatrix}. \tag{41}$$

IV. Cost Function of Calibration Problem

The very intuitive optimization problem we encounter during calibration process is given by

$$\min_{\mathbf{C},\mathbf{Z}} \|\mathbf{CX} - \mathbf{AZ}\|_F^2, \tag{42}$$

where $\mathbf{C}$ is the unknown calibration matrix, $\mathbf{X}$ is the observation matrix with each column as a BV corresponding to a specific known angle, $\mathbf{A}$ is the steering matrix with each column as the ideal steering vector corresponding to the specific known angle, and $\mathbf{Z}$ is an unknown diagonal matrix accounting for the different complex-amplitude responses (initial phase and magnitude) across the swept angles. Without any restrictions, this optimization problem will have $\mathbf{C} = \mathbf{0}$ and $\mathbf{Z} = \mathbf{0}$ as the solution which is obviously not what we want. Thus, in practice we usually impose a normalization to either $\mathbf{C}$ or $\mathbf{Z}$ to force it away from the "zero" solution and remove the multiplication ambiguity. A widely used strategy is to keep the first-row-first-column element of $\mathbf{C}$ as 1 during optimization, and the actual problem becomes

$$\begin{aligned} \min_{\mathbf{C},\mathbf{Z}} \quad & \|\mathbf{CX} - \mathbf{AZ}\|_F^2 \\ \text{s.t.} \quad & [\mathbf{C}]_{1,1} = 1 \end{aligned}. \tag{43}$$

Another normalization strategy is to impose $\|\mathbf{z}\| = 1$ where $\mathbf{z} = \mathrm{diag}\{\mathbf{Z}\}$ is the vector containing diagonal elements of $\mathbf{Z}$. The resulting optimization problem is

$$\begin{aligned} \min_{\mathbf{C},\mathbf{Z}} \quad & \|\mathbf{CX} - \mathbf{AZ}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{z}\| = 1 \end{aligned}. \tag{44}$$

We find that it is hard to further simplify the cost function in (43) using an effective way, while the cost function in (44) can be manipulated into a formulation where $\mathbf{z}$ becomes the only unknown. Note that the solution of (44) is not unique. A new solution can always be generated by multiplying the current solution with any complex number with unit magnitude. This feature is shared by eigendecomposition problem where the eigenvectors are not unique but all considered numerically accurate [1], [2]. We will show later that the nonuniqueness of the solution of (44) stems from the eigendecomposition step involved in the optimization. However, it is worth clarifying that this nonuniqueness will not affect the validity of the calibration and further angle estimation procedure since multiplying by a common complex number will not affect the mutual coupling and channel imbalance information in the calibration matrix or spatial information in the calibrated BV. Therefore, in this work we will derive the results based on formulation (44) and leave the nonuniqueness to the eigendecomposition solver embedded in our solution [3].

A. Further simplification of (44)

Suppose $\mathbf{Z}$ is known in (44), and the least squares solution of $\mathbf{C}$ will be

$$\hat{\mathbf{C}} = \mathbf{A}\mathbf{Z}\mathbf{X}^H \left(\mathbf{X}\mathbf{X}^H\right)^{-1}. \tag{45}$$

By substituting (45) into (44), we get the following simplified formulation

$$\begin{aligned} \min_{\mathbf{Z}} \quad & \|\mathbf{A}\mathbf{Z}\mathbf{P}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{z}\| = 1 \end{aligned}, \tag{46}$$

where $\mathbf{P} = \mathbf{I}_T - \mathbf{X}^H \left(\mathbf{X}\mathbf{X}^H\right)^{-1} \mathbf{X}$. This cost function actually evaluates how well the steering matrix from the array can fit in the subspace formed by raw data observations. By noting that $\mathbf{Z}$ is a diagonal matrix, we can further vectorize (46) into

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{Q}\mathbf{z}\|^2 \\ \text{s.t.} \quad & \|\mathbf{z}\| = 1 \end{aligned}, \tag{47}$$

where

$$\mathbf{Q} = \begin{bmatrix} \mathbf{A} \cdot \mathrm{diag}\{\mathbf{P}_1\} \\ \mathbf{A} \cdot \mathrm{diag}\{\mathbf{P}_2\} \\ \vdots \\ \mathbf{A} \cdot \mathrm{diag}\{\mathbf{P}_T\} \end{bmatrix}, \tag{48}$$

and $\mathbf{P}_t$ stands for the t-th column of matrix $\mathbf{P}$. Further putting it into Rayleigh quotient form, we obtain

$$\begin{aligned} \min_{\mathbf{z}} \quad & \mathbf{z}^H \mathbf{W} \mathbf{z} \\ \text{s.t.} \quad & \|\mathbf{z}\| = 1 \end{aligned}, \tag{49}$$

where $\mathbf{W} = \mathbf{Q}^H\mathbf{Q} = \left(\mathbf{A}^H\mathbf{A}\right) \odot \left(\overline{\mathbf{P}}\mathbf{P}^T\right)$ and $\overline{\mathbf{P}}$ stands for the complex conjugate of matrix $\mathbf{P}$. Clearly, the minimum is achieved at the smallest eigenvalue of matrix $\mathbf{W}$, i.e., $\sigma_0\{\mathbf{W}\}$, and the solution of $\mathbf{z}$ is $\mathbf{e}_0\{\mathbf{W}\}$ which is the normalized eigenvector associated with $\sigma_0\{\mathbf{W}\}$ [4]–[6]. Consequently, calibration matrix can be obtained using (45).

B. Involving misalignment unknowns

The above derivations are based on the condition that the steering matrix $\mathbf{A}$ is fully known, which is not true when misalignment happens during calibration data collection. In the near-field scenario, the steering matrix is given by $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_T] \in \mathbb{C}^{MN \times T}$ where

$$\mathbf{a}_t = [\phi_{1,1,t}, \phi_{1,2,t}, \ldots, \phi_{1,N,t}, \phi_{2,1,t}, \phi_{2,2,t}, \ldots, \phi_{2,N,t}, \ldots, \phi_{M,1,t}, \phi_{M,2,t}, \ldots, \phi_{M,N,t}]^T \tag{50}$$

for $t = 1, 2, ..., T$. According to the discussion in Section II, the steering matrix will be a function of the misalignment unknowns, i.e., $\mathbf{A}(\boldsymbol{\nu})$. By taking misalignment into account, the calibration problem becomes

$$\min_{\boldsymbol{\nu}} \quad \sigma_0 \left\{ \mathbf{W}(\boldsymbol{\nu}) \right\}, \tag{51}$$

where $\mathbf{W}(\boldsymbol{\nu}) = \mathbf{S}(\boldsymbol{\nu}) \odot \mathbf{U}$ with $\mathbf{S}(\boldsymbol{\nu}) = \mathbf{A}(\boldsymbol{\nu})^H \mathbf{A}(\boldsymbol{\nu})$ and $\mathbf{U} = \overline{\mathbf{P}}\mathbf{P}^T$. For $i = 1, 2, ..., T$ and $j = 1, 2, ..., T$, we have

$$[\mathbf{S}(\boldsymbol{\nu})]_{i,j} = \mathbf{a}_i^H \mathbf{a}_j = \sum_{m=1}^{M} \sum_{n=1}^{N} \phi_{m,n,i}^* \phi_{m,n,j} = \sum_{m=1}^{M} \sum_{n=1}^{N} \exp\left\{ \jmath 2\pi \frac{d_{m,n,j} - d_{m,n,i}}{\lambda} \right\}. \tag{52}$$

V. Misalignment Estimation

It is obvious that (51) does not have a closed-form solution. One straightforward approach is to update the unknowns alternatively in each iteration (see the flow chart in Fig. 6), e.g., pitch, yaw, and rotation center offset are fixed at the values from the last iteration when updating roll in the current iteration, and the iterations stop once it is converged. This approach works well from the accuracy point of view but suffers from low computational efficiency. Another way is to use iterative

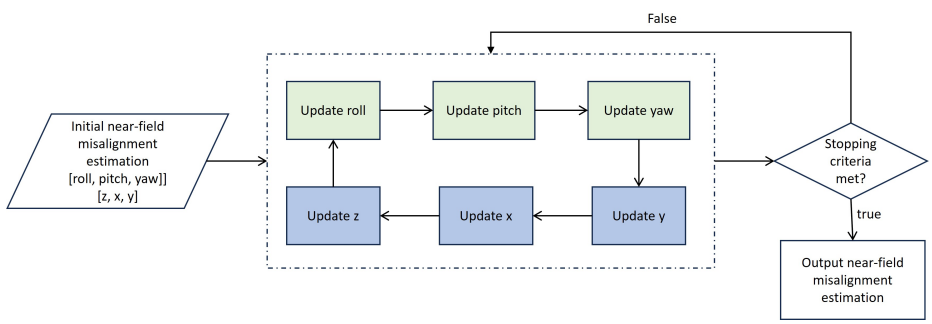

Fig. 6: Flow chart of method that updates the unknown variables alternatively.

numerical updating to find the solution. Many ready-to-use nonlinear minimization solvers can handle this problem [7], [8], and most of them will calculate the approximated gradient vector and Hessian matrix numerically in each iteration, which is very time consuming. Here we derive the exact gradient vector and Hessian matrix for the problem (51) and input them to the solver so that the computation complexity is significantly reduced. For instance, by denoting the cost function as $f(\boldsymbol{\nu}) = \sigma_0 \left\{ \mathbf{W}(\boldsymbol{\nu}) \right\}$, the gradient vector $\mathbf{g}(\boldsymbol{\nu}) = \frac{\partial f(\boldsymbol{\nu})}{\partial \boldsymbol{\nu}} \in \mathbb{R}^6$, and the Hessian matrix $\mathbf{H}(\boldsymbol{\nu}) = \frac{\partial^2 f(\boldsymbol{\nu})}{\partial \boldsymbol{\nu}^2} \in \mathbb{R}^{6\times 6}$, the updating in Newton's method can be written as [7]

$$\boldsymbol{\nu}_{i+1} = \boldsymbol{\nu}_i - \mu_i \mathbf{H}(\boldsymbol{\nu}_i)^{-1} \mathbf{g}(\boldsymbol{\nu}_i), \tag{53}$$

where μ_i is the step size such that $f(\boldsymbol{\nu}_{i+1}) < f(\boldsymbol{\nu}_i)$. Other solvers residing in the spectrum of trust-region method can also be employed when gradient and Hessian are readily calculated [9]–[13].

A. Calculation of elements in gradient vector

For $k = 1, 2, \ldots, 6$, we have [14], [15]

$$\frac{\partial f(\boldsymbol{\nu})}{\partial \nu_k} = \mathbf{e}_0^H \left\{\mathbf{W}(\boldsymbol{\nu})\right\} \frac{\partial \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k} \mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}, \tag{54}$$

where $\mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}$ is the normalized eigenvector associated with $\sigma_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}$. In addition, we have

$$\frac{\partial \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k} = \frac{\partial \mathbf{S}(\boldsymbol{\nu})}{\partial \nu_k} \odot \mathbf{U} \tag{55}$$

and

$$\frac{\partial \left[\mathbf{S}(\boldsymbol{\nu})\right]_{i,j}}{\partial \nu_k} = \frac{\jmath 2\pi}{\lambda} \sum_{m=1}^{M} \sum_{n=1}^{N} \exp\left\{\jmath 2\pi \frac{d_{m,n,j} - d_{m,n,i}}{\lambda}\right\} \left(\frac{\partial d_{m,n,j}}{\partial \nu_k} - \frac{\partial d_{m,n,i}}{\partial \nu_k}\right) \tag{56}$$

B. Calculation of elements in Hessian matrix

Then, for $k = 1, 2, \ldots, 6$ and $l = 1, 2, \ldots, 6$, we have

$$\begin{aligned} \frac{\partial^2 \left[\mathbf{S}(\boldsymbol{\nu})\right]_{i,j}}{\partial \nu_k \partial \nu_l} =& \frac{\jmath 2\pi}{\lambda} \sum_{m=1}^{M} \sum_{n=1}^{N} \exp\left\{\jmath 2\pi \frac{d_{m,n,j} - d_{m,n,i}}{\lambda}\right\} \left[\left(\frac{\partial^2 d_{m,n,j}}{\partial \nu_k \partial \nu_l} - \frac{\partial^2 d_{m,n,i}}{\partial \nu_k \partial \nu_l}\right) + \right. \\ & \left. \frac{\jmath 2\pi}{\lambda} \left(\frac{\partial d_{m,n,j}}{\partial \nu_l} - \frac{\partial d_{m,n,i}}{\partial \nu_l}\right) \left(\frac{\partial d_{m,n,j}}{\partial \nu_k} - \frac{\partial d_{m,n,i}}{\partial \nu_k}\right)\right] \end{aligned} \tag{57}$$

and

$$\frac{\partial^2 \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k \partial \nu_l} = \frac{\partial^2 \mathbf{S}(\boldsymbol{\nu})}{\partial \nu_k \partial \nu_l} \odot \mathbf{U}. \tag{58}$$

The second partial derivatives of the cost function $f(\boldsymbol{\nu})$ are given by

$$\begin{aligned} \frac{\partial^2 f(\boldsymbol{\nu})}{\partial \nu_k \partial \nu_l} =& \mathbf{e}_0^H \left\{\mathbf{W}(\boldsymbol{\nu})\right\} \frac{\partial^2 \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k \partial \nu_l} \mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\} \\ & + \mathbf{e}_0^H \left\{\mathbf{W}(\boldsymbol{\nu})\right\} \frac{\partial \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k} \frac{\partial \mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}}{\partial \nu_l} \\ & + \frac{\partial \mathbf{e}_0^H \left\{\mathbf{W}(\boldsymbol{\nu})\right\}}{\partial \nu_l} \frac{\partial \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_k} \mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}, \end{aligned} \tag{59}$$

where, by ignoring the dependency on $\mathbf{W}(\boldsymbol{\nu})$ of the eigenvalue and eigenvector,

$$\frac{\partial \mathbf{e}_0 \left\{\mathbf{W}(\boldsymbol{\nu})\right\}}{\partial \nu_l} = \mathbf{E} \left(\sigma_0 \mathbf{I} - \mathbf{E}^H \mathbf{W}(\boldsymbol{\nu}) \mathbf{E}\right)^{-1} \mathbf{E}^H \left(\frac{\partial \mathbf{W}(\boldsymbol{\nu})}{\partial \nu_l}\right) \mathbf{e}_0 \tag{60}$$

and $\mathbf{E}$ is the complementary subspace of $\mathbf{e}_0$ [14], [15].

C. Warm-start of the initial guess

The aforementioned iterative algorithms need initial guess of the unknowns to start the iterations. A warm-start of the initial guess can significantly improve the optimization speed, which can be obtained based on prior information. In our case, for example, we can use a limited number of reference sensors to characterize the chamber misalignment before any mass production. Specifically, each reference sensor will be set up in the intended chamber to collect some data, finish the calibration procedure, and estimate the misalignment variables. The average values of those misalignment estimates will be used as the warm-start point for the calibration procedure in future mass production activities. Fig. 7 shows the optimization speed comparison between the so-called "random-search" method (i.e., the same method shown in Fig. 6) and "gradient search" method (i.e., trust-region method using gradient and Hessian derived above). We can see that the "gradient search" method is much more faster with the help of prior information.

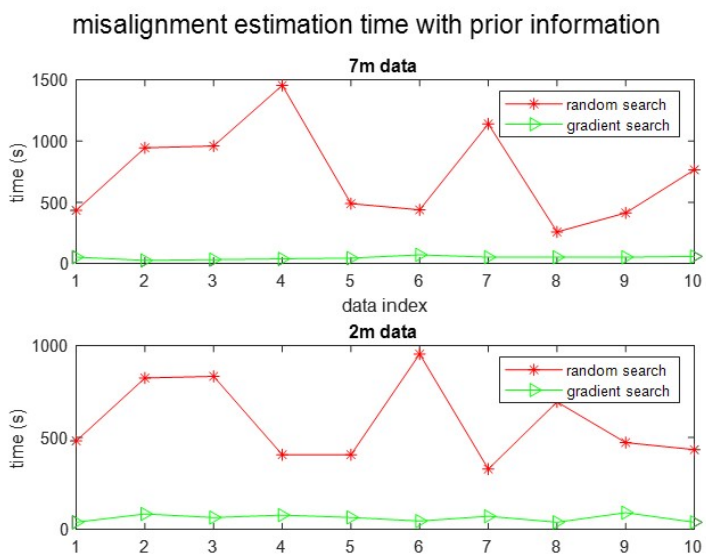

Fig. 7: Misalignment estimation time comparison with prior information.

VI. Further look into (49)

In (49), we are trying to find the smallest eigenvalue of matrix $\mathbf{W} = \left(\mathbf{A}^H\mathbf{A}\right) \odot \left(\overline{\mathbf{P}}\mathbf{P}^T\right)$. Further exploiting the projection matrix properties and elaborating the elements, we have

$$\begin{aligned}
\mathbf{W} &= \left(\mathbf{A}^H\mathbf{A}\right) \odot \overline{\mathbf{P}} \\
&= \left([\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_T]^H [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_T]\right) \odot \overline{\mathbf{P}} \\
&= \begin{bmatrix} MN & \mathbf{a}_1^H\mathbf{a}_2 & \ldots & \mathbf{a}_1^H\mathbf{a}_T \\ \mathbf{a}_2^H\mathbf{a}_1 & MN & \ldots & \mathbf{a}_2^H\mathbf{a}_T \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_T^H\mathbf{a}_1 & \mathbf{a}_T^H\mathbf{a}_2 & \ldots & MN \end{bmatrix} \odot \left(\mathbf{I}_T - \overline{\mathbf{X}^H\left(\mathbf{X}\mathbf{X}^H\right)^{-1}\mathbf{X}}\right) \\
&= MN\mathbf{I}_T - \begin{bmatrix} MN & \mathbf{a}_1^H\mathbf{a}_2 & \ldots & \mathbf{a}_1^H\mathbf{a}_T \\ \mathbf{a}_2^H\mathbf{a}_1 & MN & \ldots & \mathbf{a}_2^H\mathbf{a}_T \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_T^H\mathbf{a}_1 & \mathbf{a}_T^H\mathbf{a}_2 & \ldots & MN \end{bmatrix} \odot \overline{\mathbf{X}^H\left(\mathbf{X}\mathbf{X}^H\right)^{-1}\mathbf{X}}.
\end{aligned} \tag{61}$$

To achieve the smallest eigenvalue of $\mathbf{W}$ is equivalent to achieve the largest eigenvalue of the subtrahend. Besides, to minimize the smallest eigenvalue of $\mathbf{W}$ is equivalent to maximize the largest eigenvalue of the subtrahend.

VII. Multi-Waveform Data for Calibration

In practice, we can employ different types of waveform (e.g., different carrier frequencies, bandwidths, or multiplexing schemes) to collect chamber data for calibration. To make sure the misalignment parameters are the same across different waveform setups, we use different radar looks (frames) to convey different waveform configurations, and thus only one sweep is required. By introducing this waveform diversity and eventually more samples, the misalignment estimation will be more accurate and reliable. In the following, we present the derivations when two different types of waveform are employed, and the multi-waveform case can be easily generalized after that.

The calibration cost function for dual-waveform setup becomes

$$f(\mathbf{C}_1, \mathbf{C}_2, \mathbf{Z}_1, \mathbf{Z}_2, \boldsymbol{\nu}) = \|\mathbf{C}_1\mathbf{X}_1 - \mathbf{A}_1(\boldsymbol{\nu})\mathbf{Z}_1\|_F^2 + \|\mathbf{C}_2\mathbf{X}_2 - \mathbf{A}_2(\boldsymbol{\nu})\mathbf{Z}_2\|_F^2, \tag{62}$$

where $\mathbf{z}_1 = \mathrm{diag}\{\mathbf{Z}_1\}$, $\mathbf{z}_2 = \mathrm{diag}\{\mathbf{Z}_2\}$, $\|\mathbf{z}_1\| = 1$, and $\|\mathbf{z}_2\| = 1$. Note that we assume the calibration matrices are different for different types of waveform and the steering matrices are different due to different wavelengths (i.e., carrier frequencies are different). However, two data sets will share the

same misalignment parameters $\boldsymbol{\nu}$ because they are collected from the same sweep. The intermediate variables $\mathbf{C}_1$ and $\mathbf{C}_2$ can be estimated as

$$\mathbf{C}_1 = \mathbf{A}_1(\boldsymbol{\nu})\mathbf{Z}_1\mathbf{X}_1^H\left(\mathbf{X}_1\mathbf{X}_1^H\right)^{-1}, \tag{63}$$

and

$$\mathbf{C}_2 = \mathbf{A}_2(\boldsymbol{\nu})\mathbf{Z}_2\mathbf{X}_2^H\left(\mathbf{X}_2\mathbf{X}_2^H\right)^{-1}. \tag{64}$$

After substituting (63) and (64) into (62), we have

$$\begin{aligned} f(\mathbf{Z}_1,\mathbf{Z}_2,\boldsymbol{\nu}) =& \left\|\mathbf{A}_1(\boldsymbol{\nu})\mathbf{Z}_1\mathbf{X}_1^H\left(\mathbf{X}_1\mathbf{X}_1^H\right)^{-1}\mathbf{X}_1 - \mathbf{A}_1(\boldsymbol{\nu})\mathbf{Z}_1\right\|_F^2 \\ &+ \left\|\mathbf{A}_2(\boldsymbol{\nu})\mathbf{Z}_2\mathbf{X}_2^H\left(\mathbf{X}_2\mathbf{X}_2^H\right)^{-1}\mathbf{X}_2 - \mathbf{A}_2(\boldsymbol{\nu})\mathbf{Z}_2\right\|_F^2 \\ =& \|\mathbf{A}_1(\boldsymbol{\nu})\mathbf{Z}_1\mathbf{P}_1\|_F^2 + \|\mathbf{A}_2(\boldsymbol{\nu})\mathbf{Z}_2\mathbf{P}_2\|_F^2 \\ =& \|\mathbf{Q}_1(\boldsymbol{\nu})\mathbf{z}_1\|^2 + \|\mathbf{Q}_2(\boldsymbol{\nu})\mathbf{z}_2\|^2 \\ =& \mathbf{z}_1^H\mathbf{W}_1(\boldsymbol{\nu})\mathbf{z}_1 + \mathbf{z}_2^H\mathbf{W}_2(\boldsymbol{\nu})\mathbf{z}_2, \end{aligned} \tag{65}$$

where

$$\mathbf{P}_1 = \mathbf{I}_T - \mathbf{X}_1^H\left(\mathbf{X}_1\mathbf{X}_1^H\right)^{-1}\mathbf{X}_1, \tag{66}$$

$$\mathbf{P}_2 = \mathbf{I}_T - \mathbf{X}_2^H\left(\mathbf{X}_2\mathbf{X}_2^H\right)^{-1}\mathbf{X}_2, \tag{67}$$

$$\mathbf{W}_1(\boldsymbol{\nu}) = \mathbf{Q}_1(\boldsymbol{\nu})^H\mathbf{Q}_1(\boldsymbol{\nu}) = \left(\mathbf{A}_1(\boldsymbol{\nu})^H\mathbf{A}_1(\boldsymbol{\nu})\right)\odot\left(\overline{\mathbf{P}}_1\mathbf{P}_1^T\right), \tag{68}$$

$$\mathbf{W}_2(\boldsymbol{\nu}) = \mathbf{Q}_2(\boldsymbol{\nu})^H\mathbf{Q}_2(\boldsymbol{\nu}) = \left(\mathbf{A}_2(\boldsymbol{\nu})^H\mathbf{A}_2(\boldsymbol{\nu})\right)\odot\left(\overline{\mathbf{P}}_2\mathbf{P}_2^T\right). \tag{69}$$

Therefore, to minimize the cost function, we should solve the following problem

$$\min_{\boldsymbol{\nu}} \sigma_0\left\{\mathbf{W}_1(\boldsymbol{\nu})\right\} + \sigma_0\left\{\mathbf{W}_2(\boldsymbol{\nu})\right\}. \tag{70}$$

The numerical method to solve this problem should be a straightforward extension of the one mentioned in the last section. When we have K data sets collected from K different types of waveform, the problem will be generalized as

$$\min_{\boldsymbol{\nu}} \sum_{k=1}^{K} \sigma_0\left\{\mathbf{W}_k(\boldsymbol{\nu})\right\}. \tag{71}$$

VIII. Summary

In this document, we introduce near-field calibration problem for MIMO radar. There are several benefits for near-field calibration when it comes to mass production of large-aperture automotive radar (small-aperture radar can also enjoy these benefits). However, near-field misalignment (on both roll-pitch-yaw and rotation center offset) in the chamber is inevitable, and it can significantly downgrade the calibration performance if ignored by our data model. Therefore, when we formulate our near-field calibration problem, such misalignment is considered in the data model and assumed unknown. During the optimization procedure of the calibration problem, misalignment related unknowns are estimated and naturally compensated for so that the generated calibration matrix is not contaminated by the misalignment effects. The novel part is that we define the cost function independent of calibration matrix and find the gradient vector and Hessian matrix for a faster and better convergence in search of misalignment unknowns. In the end, we propose a coherent calibration method using multiple sets of data based on our misalignment estimation and compensation approaches.

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